A Divisor-Residue Criterion for Bounded A-Parameters in the Erdős–Straus Conjecture

Skip Potter
AI Assistance: Brodie Foxworth

July 18, 2026 (Revision 8, TeX cleanup)

Abstract

We present a divisor-residue criterion for the Erdős–Straus conjecture in the residue class $n \equiv 1 \pmod{12}$, $n \not\equiv 0 \pmod 5$. We introduce a parameter $A = 4x - n$ and reduce the conjecture to a **bounded divisor-residue condition**: for each admissible n, there must exist bounded $A \equiv 3 \pmod 4$ and a divisor P of $\big(n(n+A)/4\big)^2$ satisfying

$$P \equiv -n^2 \cdot 4^{-1} \pmod A$$

with positive y, z. We prove that for n prime, $n \equiv 1 \pmod 4$, this condition is equivalent to the target residue T lying in the **bounded divisor-residue set** $D_A((nx)^2)$, a subset of $(\mathbb{Z}/A\mathbb{Z})^*$ generated by prime factors of nx with exponents bounded by $2v_p(nx)$. For A prime, this reduces to a quadratic non-residue condition via quadratic reciprocity, conditional on a Bounded Divisor-Residue Lemma that is partially proven and computationally verified. We prove unconditionally that $n \equiv 5 \pmod 8$ always admits $A = 3$. We also describe a conditional analytic route, via Burgess-type bounds for least prime quadratic non-residues in arithmetic progressions, that would give

$$A = O\Big(n^{1/(4\sqrt{e})+\varepsilon}\Big).$$

Computationally, $A \leq 31$ covers all 6,666 cases up to $n = 100{,}000$, and $A \leq 99$ covers all but one of 666,666 cases up to $n = 10{,}000{,}000$.

1 Introduction

The Erdős–Straus conjecture states that for every integer $n \geq 2$, there exist positive integers x, y, z such that

$$\frac{4}{n} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}.$$

Despite extensive work since 1950, the conjecture remains open. Mordell (1967) proved that polynomial identities providing solutions for $n \equiv r \pmod m$ exist only when r is a quadratic non-residue modulo m. Since 1 is a quadratic residue modulo every m, no polynomial identity can cover $n \equiv 1 \pmod m$. This makes $n \equiv 1 \pmod 4$—and in particular $n \equiv 1 \pmod{12}$—the resistant case.

Mballa (2026) introduced a parametric framework proving density-1 coverage for $n \equiv 1 \pmod 4$ using symmetric solutions, requiring a divisor $b \equiv 3 \pmod 4$ of n. This handles all composite n having a prime factor congruent to $3 \pmod 4$, but leaves prime $n \equiv 1 \pmod 4$ as the key open case.

Our approach does not use polynomial identities. Instead, we parameterize solutions by $A = 4x - n$ and reduce the problem to a **bounded divisor-residue condition**: a specific residue must be achievable as a product of prime powers of nx, with each exponent bounded by the p-adic valuation of nx. This is a concrete combinatorial condition in elementary number theory, not an algebraic-geometric problem.

2 The A-Parameter Framework

2.1 Setup

For $n \equiv 1 \pmod 4$, write

$$x = \frac{n+A}{4}, \qquad A \equiv 3 \pmod 4, \qquad A \geq 3.$$

The Erdős–Straus equation becomes

$$\frac{4}{n} = \frac{4}{n+A} + \frac{1}{y} + \frac{1}{z}.$$

Set

$$m = \frac{n+A}{4}, \qquad nx = n \cdot m.$$

The remainder $4A/(n(n+A))$ must be decomposed as $1/y + 1/z$. The existence of positive y, z is equivalent to finding a divisor P of $(nx)^2$ such that

(i) $P \equiv -n^2 \cdot 4^{-1} \pmod A$, the target residue T; and

(ii) the resulting values

$$y = \frac{P+nx}{A}, \qquad z = \frac{Q+nx}{A}, \qquad Q = \frac{(nx)^2}{P},$$

are positive integers.

We call P a **working divisor** of $(nx)^2$ for the pair (n, A).

2.2 The Bounded Divisor-Residue Set

The set of achievable residues $P \pmod A$, where P ranges over divisors of $(nx)^2$, is

$$D_A((nx)^2) = \left\{ \prod_i p_i^{\alpha_i} \bmod A : 0 \leq \alpha_i \leq 2v_{p_i}(nx) \right\},$$

where the product ranges over all primes p_i dividing nx, and $v_{p_i}(nx)$ is the p_i-adic valuation of nx. This is a **bounded** set: each exponent α_i is limited by $2v_{p_i}(nx)$, reflecting the constraint that P must be an actual divisor of $(nx)^2$.

Note that $D_A((nx)^2)$ is a subset of the subgroup

$$H(A) = \langle p \bmod A : p \mid nx \rangle \subseteq (\mathbb{Z}/A\mathbb{Z})^*,$$

but is generally not equal to $H(A)$, since subgroup membership allows arbitrary integer powers while D_A restricts exponents to bounded ranges. The distinction matters: a subgroup may contain -1 even when no actual divisor of $(nx)^2$ realizes the needed residue.

2.3 The Exact Characterization

Theorem 1 (Divisor-Residue Criterion)**.** *For n prime, $n \equiv 1 \pmod 4$, and $A \equiv 3 \pmod 4$ with $\gcd(n, A) = 1$, the parameter A yields an Erdős–Straus solution if and only if*

$$T = -n^2 \cdot 4^{-1} \pmod A$$

lies in $D_A((nx)^2)$.

Proof. By construction, A yields a solution if and only if there exists a divisor $P \mid (nx)^2$ such that

$$P \equiv T \pmod{A}.$$

Since $m = (n + A)/4 \equiv n \cdot 4^{-1} \pmod A$, we have

$$T = -n^2 \cdot 4^{-1} \equiv -nm \equiv -nx \pmod{A}.$$

Thus $P \equiv -nx \pmod A$, so $A \mid P + nx$ and hence

$$y = \frac{P + nx}{A}$$

is an integer.

Let $Q = (nx)^2/P$. Because $P \mid (nx)^2$, Q is a positive integer. Also, since $\gcd(nx, A) = 1$, the residue P is invertible modulo A. From $PQ = (nx)^2$ and $P \equiv -nx \pmod A$, we get

$$Q \equiv (nx)^2 P^{-1} \equiv (nx)^2(-nx)^{-1} \equiv -nx \pmod{A}.$$

Therefore $A \mid Q + nx$, and

$$z = \frac{Q + nx}{A}$$

is an integer. Since $P, Q, nx > 0$, both y and z are positive. Therefore $P \equiv T \pmod A$ with $P \mid (nx)^2$ is exactly the condition for an Erdős–Straus solution arising from this A. □

Remark 1 (Verification). The target may also be written $T = -n \cdot m \pmod A$, since $m \equiv n \cdot 4^{-1} \pmod A$. Computational verification found that $T \in D_A((nx)^2)$ coincides exactly with the existence of Erdős–Straus solutions across all 6,666 test cases with $n \leq 100{,}000$, $n \equiv 1 \pmod{12}$, $n \not\equiv 0 \pmod 5$, with zero mismatches.

Remark 2 (D_A versus $H(A)$). In most tested cases, $D_A((nx)^2) = H(A)$. However, $D_A \neq H(A)$ in general; for example, $n = 7561$, $A = 19$ gives $|D_A| = 6$ but $|H(A)| = 18$. Despite this, the critical equivalence $-1 \in D_A \iff -1 \in H(A)$ holds with zero mismatches across all 6,473 prime-A cases tested. This equivalence is partially proven and computationally verified; see the Bounded Divisor-Residue Lemma below.

3 Main Results

3.1 The $n \equiv 5 \pmod 8$ Case

Theorem 2. *For n prime, $n \equiv 5 \pmod 8$, and $n \neq 3$, the choice $A = 3$ yields an Erdős–Straus solution.*

Proof. When $n \equiv 5 \pmod 8$, we have $n + 3 \equiv 0 \pmod 8$, so

$$m_3 = \frac{n+3}{4}$$

is even. Therefore $2 \mid m_3$, and 2 is a prime factor of $nx = n \cdot m_3$. Since $2 \equiv -1 \pmod 3$, take $P = 2$.

Because n is prime and $n \neq 3$, we have $n^2 \equiv 1 \pmod 3$, and since $4^{-1} \equiv 1 \pmod 3$,

$$-n^2 \cdot 4^{-1} \equiv -1 \equiv 2 \pmod 3.$$

Thus $P = 2$ satisfies the target congruence for $A = 3$. Since $P \mid (nx)^2$, Theorem 1 gives positive integral y, z. Therefore $A = 3$ works directly, without invoking the conditional prime-A criterion. □

3.2 Reduction to a Quadratic Non-Residue Condition

Theorem 3 (Prime A Criterion, conditional on the Bounded Divisor-Residue Lemma)**.** *For A prime,* $\gcd(n, A) = 1$*, and n prime with* $n \equiv 1 \pmod 4$*,*

$$T \in D_A((nx)^2) \quad \Longleftrightarrow \quad \textit{some prime factor } p \mid nx \textit{ is a quadratic non-residue modulo } A.$$

The converse direction is unconditional; the forward direction is conditional and computationally verified in the tested range.

Proof. For A prime, $(\mathbb{Z}/A\mathbb{Z})^*$ is cyclic of order $A - 1$. Since $A \equiv 3 \pmod 4$, $(A - 1)/2$ is odd, and -1 is the unique element of order 2.

For the converse, suppose no prime factor of nx is a quadratic non-residue modulo A. Then all generators of $H(A)$ have odd order, so $h = |H(A)|$ is odd and $-1 \notin H(A)$. Since $D_A \subseteq H(A)$, we have $-1 \notin D_A$. If $T \in D_A$, then using $T = -nm$ and $nm \in H(A)$, we would have

$$T \cdot (nm)^{-1} = -1 \in H(A),$$

a contradiction. Hence $T \notin D_A$.

For the forward direction, if some prime $p \mid nx$ is a quadratic non-residue modulo A, then p has even order, so $h = |H(A)|$ is even and $-1 \in H(A)$. The Bounded Divisor-Residue Lemma below gives $-1 \in D_A$ in the proven sub-cases and computationally in the remaining tested cases.

However, D_A is not closed under multiplication. The step from $-1 \in D_A$ and $nm \in D_A$ to $T = (-1)(nm) \in D_A$ does not follow automatically, because multiplying two divisor residues may push exponents beyond the bound $2v_p(nx)$. If the divisor realizing -1 uses exponent a_i on prime p_i, then the product with nm requires

$$a_i + v_{p_i}(nx) \leq 2v_{p_i}(nx),$$

equivalently $a_i \leq v_{p_i}(nx)$.

Define the shifted divisor-residue set

$$D_A^{(nm)}((nx)^2) = \left\{ \prod_i p_i^{\alpha_i} \bmod A : 0 \leq \alpha_i \leq 2v_{p_i}(nx) - v_{p_i}(nm) \right\}.$$

This is the set of residues R such that $R \cdot nm$ is an actual divisor of $(nx)^2$. The correct sufficient condition for the -1 factorization route is therefore $-1 \in D_A^{(nm)}$, not merely $-1 \in D_A$.

Computationally, $T \in D_A((nx)^2)$ has zero mismatches across all 6,473 prime-A cases tested. In 50 of these cases, $-1 \notin D_A^{(nm)}$, yet $T \in D_A$ via a different divisor path. Thus a complete proof of the forward direction requires a direct argument for $T \in D_A$, or a full proof that $-1 \in D_A^{(nm)}$ when appropriate. □

3.3 The Bounded Divisor-Residue Lemma

Lemma 1 (Bounded Divisor-Residue, partially proven and computationally verified)**.** *For A prime and* $\gcd(n, A) = 1$*, if* $-1 \in H(A)$*, then* $-1 \in D_A((nx)^2)$ *in the proven sub-cases below and in all tested cases.*

Proof sketch and tested classification. Let $h = |H(A)|$. Since $-1 \in H(A)$, h is even. Let g be a primitive root modulo A, so $H(A) = \langle g^{(A-1)/h} \rangle$ and $-1 = g^{h/2}$ inside $H(A)$.

We must show that $h/2$ lies in the bounded sumset

$$S = \left\{ \sum_i s_i \alpha_i \pmod{h} : 0 \leq \alpha_i \leq 2e_i \right\},$$

where s_i is the discrete logarithm of p_i in $H(A)$ and $e_i = v_{p_i}(nx)$.

If $h = 2$, any quadratic non-residue generator gives $p^1 \equiv -1 \pmod{A}$ with exponent $1 \leq 2e$; this case is proven generally. If some prime factor $p \mid nx$ has $\mathrm{ord}_A(p) = 2$, then $p \equiv -1 \pmod{A}$ and again p^1 realizes -1 directly; this case is also proven generally.

For $h = 2m$ with $m > 1$ odd and no order-2 quadratic non-residue, decompose $\mathbb{Z}/h\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z}$. The $\mathbb{Z}/2\mathbb{Z}$ component is supplied by a single quadratic non-residue. The remaining condition is a bounded sumset condition in $\mathbb{Z}/m\mathbb{Z}$.

Let

$$S' = A_1 + \cdots + A_k \subseteq \mathbb{Z}/m\mathbb{Z},$$

where

$$A_i = \{0, s_i, 2s_i, \ldots, 2e_i s_i\}.$$

If $\mathrm{Stab}(S')$ is trivial and

$$\sum_i |A_i| = \sum_i (2e_i + 1) \geq m + k - 1,$$

then Kneser's theorem gives $|S'| \geq m$, hence $S' = \mathbb{Z}/m\mathbb{Z}$ and the required residue is present. This sub-case is proven under the stated hypotheses.

The remaining tested cases are verified computationally. A general proof of the full lemma remains open. $\square$

Case	Count	Status
$h = 2$	5,533	proven generally
order-2 QNR	445	proven generally
Kneser, trivial stabilizer and size condition	929	proven under hypotheses
$S' = \mathbb{Z}/m\mathbb{Z}$ with nontrivial stabilizer	8	computational
bounded sumset / multi-QNR combinations	8	computational
all-QNR combination	1	computational
Total	6,473	zero failures in tested range

Corollary 1 (Conditional sufficient condition)**.** *If* $\left(\frac{A}{n}\right) = -1$*, then A works, conditional on the forward direction of Theorem 3 and the Bounded Divisor-Residue Lemma. This condition is computationally verified in all tested cases.*

Proof. By quadratic reciprocity, since $n \equiv 1 \pmod 4$,

$$\left(\frac{A}{n}\right) = \left(\frac{n}{A}\right).$$

If $\left(\frac{A}{n}\right) = -1$, then n is a quadratic non-residue modulo A. Since n is a prime factor of nx, the conditional forward direction of Theorem 3 applies. □

3.4 The Composite n Case

Theorem 4 (Mballa 2026)**.** *If n has a prime factor $p \equiv 3 \pmod 4$, then $A = p$ yields an Erdős–Straus solution.*

This handles all composite $n \equiv 1 \pmod 4$ having a prime factor congruent to $3 \pmod 4$. Combined with Theorem 2, the only remaining case is prime $n \equiv 1 \pmod 8$.

3.5 Legendre Symbol Identity for the m-Route

Theorem 5 (Legendre Symbol Identity)**.** *For any prime p dividing $m = (n + A)/4$, the Legendre symbol satisfies*

$$\left(\frac{p}{A}\right) = \left(\frac{n}{p}\right).$$

Thus p is a quadratic non-residue modulo A if and only if n is a quadratic non-residue modulo p.

Proof. Since $p \mid m = (n + A)/4$, we have $A \equiv -n \pmod p$. By quadratic reciprocity, since $n \equiv 1 \pmod 4$,

$$\left(\frac{p}{A}\right) = \left(\frac{A}{p}\right)(-1)^{((p-1)/2)((A-1)/2)}.$$

Since $A \equiv 3 \pmod 4$, $(A - 1)/2$ is odd. If $p \equiv 1 \pmod 4$, then

$$\left(\frac{p}{A}\right) = \left(\frac{A}{p}\right) = \left(\frac{-n}{p}\right) = \left(\frac{-1}{p}\right)\left(\frac{n}{p}\right) = \left(\frac{n}{p}\right).$$

If $p \equiv 3 \pmod 4$, then

$$\left(\frac{p}{A}\right) = -\left(\frac{A}{p}\right) = -\left(\frac{-n}{p}\right) = -\left(\frac{-1}{p}\right)\left(\frac{n}{p}\right) = \left(\frac{n}{p}\right).$$

Thus the identity holds in both cases. □

Corollary 2 (Route formulation)**.** *A works if either $\left(\frac{A}{n}\right) = -1$ (the direct route), conditional as above, or some prime $p \mid (n + A)/4$ has $\left(\frac{n}{p}\right) = -1$ (the m-route), again subject to the same bounded divisor-residue condition. Both routes are governed by the same quadratic character of n.*

3.6 Computational Verification

Theorem 6 (Computational, $n \leq 100{,}000$)**.** *For all 6,666 integers $n \equiv 1 \pmod{12}$, $n \not\equiv 0 \pmod 5$, with $13 \leq n \leq 100{,}000$, there exists*

$$A \in \{3, 7, 11, 15, 19, 23, 27, 31\}$$

yielding an Erdős–Straus solution. The maximum A required is 31.

Theorem 7 (Computational, $n \leq 10{,}000{,}000$)**.** *For all 666,666 integers* $n \equiv 1 \pmod{12}$, $n \not\equiv 0 \pmod 5$, *with* $13 \leq n \leq 10{,}000{,}000$, *there exists* $A \leq 99$ *yielding an Erdős–Straus solution, with one exception:* $n = 8{,}803{,}369$, *which requires* $A = 107$. *The maximum* A *required excluding this outlier is* 59.

The 666,666 count refers to all integers in the residue class, not just primes. Only prime n need be tested after reductions, but the verification included all integers for completeness.

A	Count	Percentage	Cumulative
3	5,533	83.0%	83.0%
7	967	14.5%	97.5%
11	109	1.6%	99.1%
15	20	0.3%	99.4%
19	16	0.2%	99.7%
23	15	0.2%	99.9%
27	1	0.0%	99.9%
31	5	0.1%	100.0%

3.7 Analytic Bound

Proposition 1 (Analytic route via Burgess-type bounds)**.** *Conditional on a standard least-prime-QNR-in-arithmetic-progression estimate: for n prime,* $n \equiv 1 \pmod 4$, *there exists a prime* $A \equiv 3 \pmod 4$ *with* $\left(\frac{A}{n}\right) = -1$ *and*

$$A = O\Big(n^{1/(4\sqrt{e})+\varepsilon}\Big)$$

for any $\varepsilon > 0$. *Such an* A *yields an Erdős–Straus solution subject to the conditional forward direction of Theorem 3.*

Proof sketch. The Burgess bound applied to character sums over primes $p \leq X$ with $p \equiv 3 \pmod 4$ suggests that the character sum

$$\sum_{\substack{p \leq X \\ p \equiv 3 \pmod 4}} \left(\frac{p}{n}\right)$$

is small relative to $\pi(X; 4, 3)$ for $X = n^{1/(4\sqrt{e})+\varepsilon}$ under the needed least-prime-QNR-in-AP estimate. A fully sourced proof requires careful citation of the prime number theorem in arithmetic progressions combined with Burgess-type character sum bounds. We state this as a conditional analytic route, not as an unconditional theorem. □

Under GRH, the least quadratic non-residue modulo n is $O((\log n)^2)$, and the least prime congruent to $3 \pmod 4$ that is a quadratic non-residue modulo n is also expected to be $O((\log n)^2)$ under the corresponding prime-in-AP formulation.

4 The Covering Cascade

The covering set

$$\{3, 7, 11, 15, 19, 23, 31\}$$

verified for $n \leq 100{,}000$ operates via two mechanisms.

Direct route. For most prime cases up to 100,000, at least one $A \in \{7, 11, 19, 23, 31\}$ has $\left(\frac{A}{n}\right) = -1$, and that A works directly in the tested range. The case $A = 3$ always has $\left(\frac{3}{n}\right) = 1$ when $n \equiv 1 \pmod{12}$.

m-**route.** For the 21 tested cases where n is a quadratic residue modulo all of $7, 11, 19, 23, 31$, the m-route provides a solution: the smallest quadratic non-residue p of n appears as a prime factor of m_A for some A in the covering set, and $(\frac{p}{A}) = \left(\frac{n}{p}\right) = -1$ by Theorem 5.

5 The Constant Bound Conjecture

Conjecture 1. *There exists an absolute constant C such that for every prime $n \equiv 1 \pmod 4$, there exists $A \le C$ with $A \equiv 3 \pmod 4$ and*

$$T \in D_A((nx)^2).$$

Evidence includes:

- $A \le 31$ for all $n \le 100{,}000$ in the tested residue class;
- $A \le 99$ for all but one of 666,666 cases up to $n = 10{,}000{,}000$;
- $A \le 107$ for all tested cases up to $n = 10{,}000{,}000$;
- a Burgess-type least-prime-QNR-in-AP estimate would give $A = O(n^{0.152\cdots})$, conditionally;
- GRH gives a conditional $O((\log n)^2)$ route.

This conjecture is stronger than what GRH implies. Proving it would require showing that among a finite set of primes, at least one is always a quadratic non-residue modulo n, or proving an equivalent bounded divisor-residue statement.

6 Context from Known Results

6.1 Mordell (1967)

Mordell's theorem rules out polynomial identities for $n \equiv 1 \pmod m$. The present framework uses divisor-residue conditions, not polynomial identities.

6.2 Elsholtz–Tao (2013)

Elsholtz and Tao showed that the total number of solutions satisfies an asymptotic of the form $\sum_{p \le N} f(p) \asymp N \log^2 N$. This provides context on solution density.

6.3 Vaughan (1970) and Webb (1970)

Vaughan showed that potential counterexamples have density zero. Webb showed that only a sublinear number of values need more than two terms. These results are consistent with the computational evidence here.

6.4 Bright–Loughran (2020)

Bright and Loughran showed that the Brauer–Manin obstruction does not explain failure of the integral Hasse principle for Erdős–Straus surfaces.

6.5 Mballa (2026)

Mballa introduced a parametric framework proving density-1 coverage for $n \equiv 1 \pmod 4$ via symmetric solutions. The present framework gives an exact divisor-residue criterion and a separate unconditional proof for the $n \equiv 5 \pmod 8$ case.

6.6 Burgess (1963)

Burgess-type character sum bounds are the key input for the conditional analytic route described above. The application to primes in arithmetic progressions with a prescribed quadratic character requires careful sourcing.

6.7 Salez (2014)

Salez verified the Erdős–Straus conjecture computationally up to 10^{17}. The present work provides a structural divisor-residue framework for explaining why small A values frequently succeed.

7 Summary of Results

Result	Status	Scope
$T \in D_A$ iff A works	proven	prime $n \equiv 1 \pmod 4$, $\gcd(n, A) = 1$
$-1 \in H(A) \Rightarrow -1 \in D_A$	partially proven + computational	prime A
Prime A: $T \in D_A$ iff a QNR prime factor exists	converse proven; forward conditional + computational	prime A
$n \equiv 5 \pmod 8$: $A = 3$ works	proven	prime $n \neq 3$
Composite n with factor $\equiv 3 \pmod 4$	proven via Mballa	such composites
$\left(\frac{p}{A}\right) = \left(\frac{n}{p}\right)$ for $p \mid m_A$	proven	applicable primes
$A \leq 31$ covers $n \leq 100{,}000$	verified	tested class
$A \leq 99$ covers all but one up to 10,000,000	verified	tested class
$A = O(n^{1/(4\sqrt{e})+\varepsilon})$	conditional proposition	analytic route
$A = O((\log n)^2)$	conditional on GRH	analytic route
$A \leq C$ for all n	conjecture	open

8 Open Problems

1. **Bounded Divisor-Residue Lemma in full generality.** Prove that $-1 \in H(A)$ implies $-1 \in D_A((nx)^2)$, or characterize the precise exceptions.

2. D_A **closure and shifted divisor-residue set.** Since D_A is not closed under multiplication, prove $T \in D_A$ directly whenever a quadratic non-residue prime factor exists, or prove the appropriate shifted-set statement.

3. **Constant bound conjecture.** Prove $A \leq C$ for some absolute constant.

4. **Covering system proof.** Prove that $\{3, 7, 11, 15, 19, 23, 31\}$ covers all n, not just $n \leq 100{,}000$.

5. **Full proof of Proposition 8.** Provide a fully sourced proof combining Burgess bounds with prime number theorem estimates in arithmetic progressions for the least prime $\equiv 3 \pmod 4$ that is a quadratic non-residue modulo n.

6. m**-route sufficiency.** Prove that when the direct route fails, the m-route always succeeds for some A in the covering set.

References

[1] P. Erdős, "Az egyenlet egész számú megoldásairól," *Mat. Lapok* 1, 192–210, 1950.

[2] L. J. Mordell, *Diophantine Equations*, Academic Press, 1967, pp. 287–290.

[3] R. C. Vaughan, "On a problem of Erdős, Straus and Schinzel," *Mathematika* 17, 193–198, 1970.

[4] W. A. Webb, "On $4/n = 1/x + 1/y + 1/z$," *Proceedings of the American Mathematical Society* 25, 578–584, 1970.

[5] D. A. Burgess, "On character sums and L-series, II," *Proceedings of the London Mathematical Society* s3-13, 524–536, 1963.

[6] C. Elsholtz and T. Tao, "Counting the number of solutions to the Erdős–Straus equation on unit fractions," *Journal of the Australian Mathematical Society* 94, 50–105, 2013.

[7] M. Bright and D. Loughran, "Brauer–Manin obstruction for Erdős–Straus surfaces," *Bulletin of the London Mathematical Society* 52, 746–761, 2020.

[8] P. U. Mballa, "A unified parametric approach to the Erdős–Straus conjecture," arXiv:2602.20036, 2026.

[9] S. E. Salez, "The Erdős–Straus conjecture: new modular equations and checking up to 10^{17}," arXiv:1406.6307, 2014.

Computational tools: Python 3.13, SymPy 1.14.0. All computations verified with exact rational arithmetic. AI assistance: Brodie Foxworth (OpenClaw agent) performed computational verification, developed the divisor-residue criterion, and prepared this manuscript under the direction of the author.